

Scalability & Future Plan

Project Information

Date	15 March 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the current project solution, **OptiCrop: Smart Agricultural Production Optimization Engine**, can be scaled to handle larger user bases, increased agricultural data loads, and extended smart farming features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current completed state.

Current System Limitations

S.No	Limitation	Impact	Priority
1	The system currently uses manually entered soil and weather parameters.	Real-time farm condition monitoring is not available without external sensors or APIs.	High
2	Data storage uses SQLite for prediction history and records.	Performance may reduce when handling large numbers of users and prediction records.	Medium
3	The model is trained on a fixed crop recommendation dataset.	Predictions may need improvement for region-specific crops, soil types, and seasonal patterns.	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask application deployed for standard project demonstration.	Deploy the application on Render or cloud infrastructure with containerization and load balancing.
2	Data Storage	SQLite database for storing predictions and history.	Migrate to PostgreSQL or MySQL for better concurrency, reliability, and large-scale data handling.
3	Prediction Accuracy	Random Forest model trained on available crop recommendation dataset.	Retrain the model using larger regional datasets, seasonal data, and farmer feedback records.
4	Smart Farming Integration	Manual input of soil nutrients and environmental values.	Integrate Weather API, IoT soil sensors, and satellite data for automated and real-time input collection.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Time-line	Expected Impact
Phase 1	Integrate Weather API and improve dashboard analytics.	Q3 2026	Provide more accurate recommendations using real-time environmental data.
Phase 2	Migrate from SQLite to PostgreSQL and add secure user authentication.	Q4 2026	Support multiple users, secure prediction history, and scalable database operations.
Phase 3	Add IoT sensor support, disease prediction, fertilizer recommendation AI, and yield prediction.	Q1 2027	Transform OptiCrop into a complete smart agriculture decision-support platform.